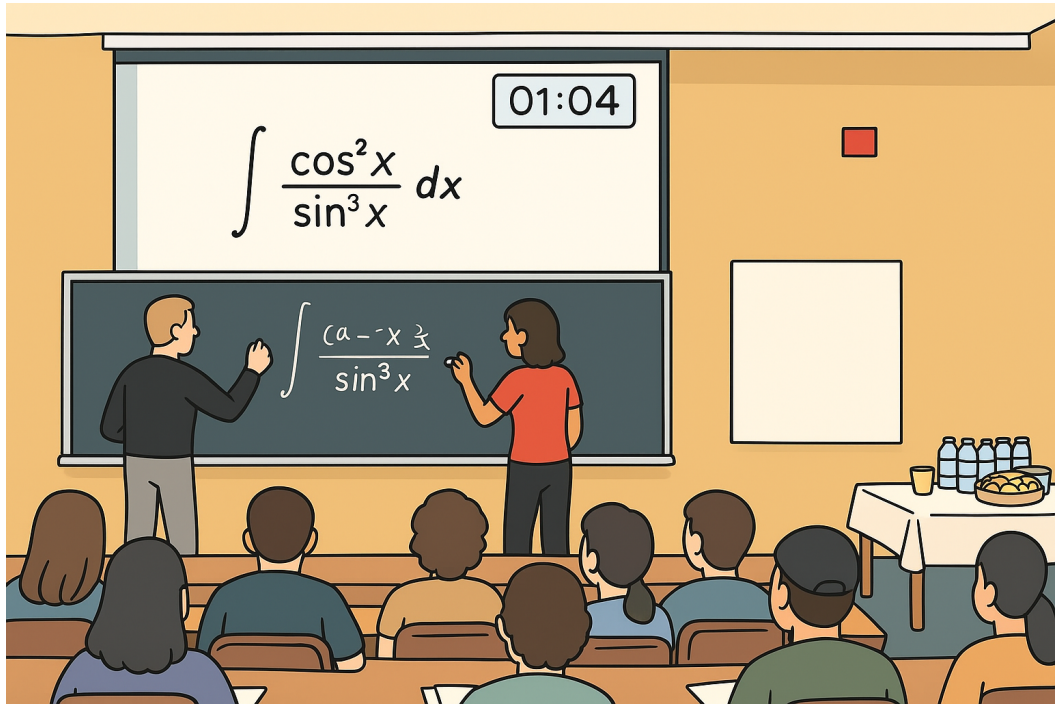
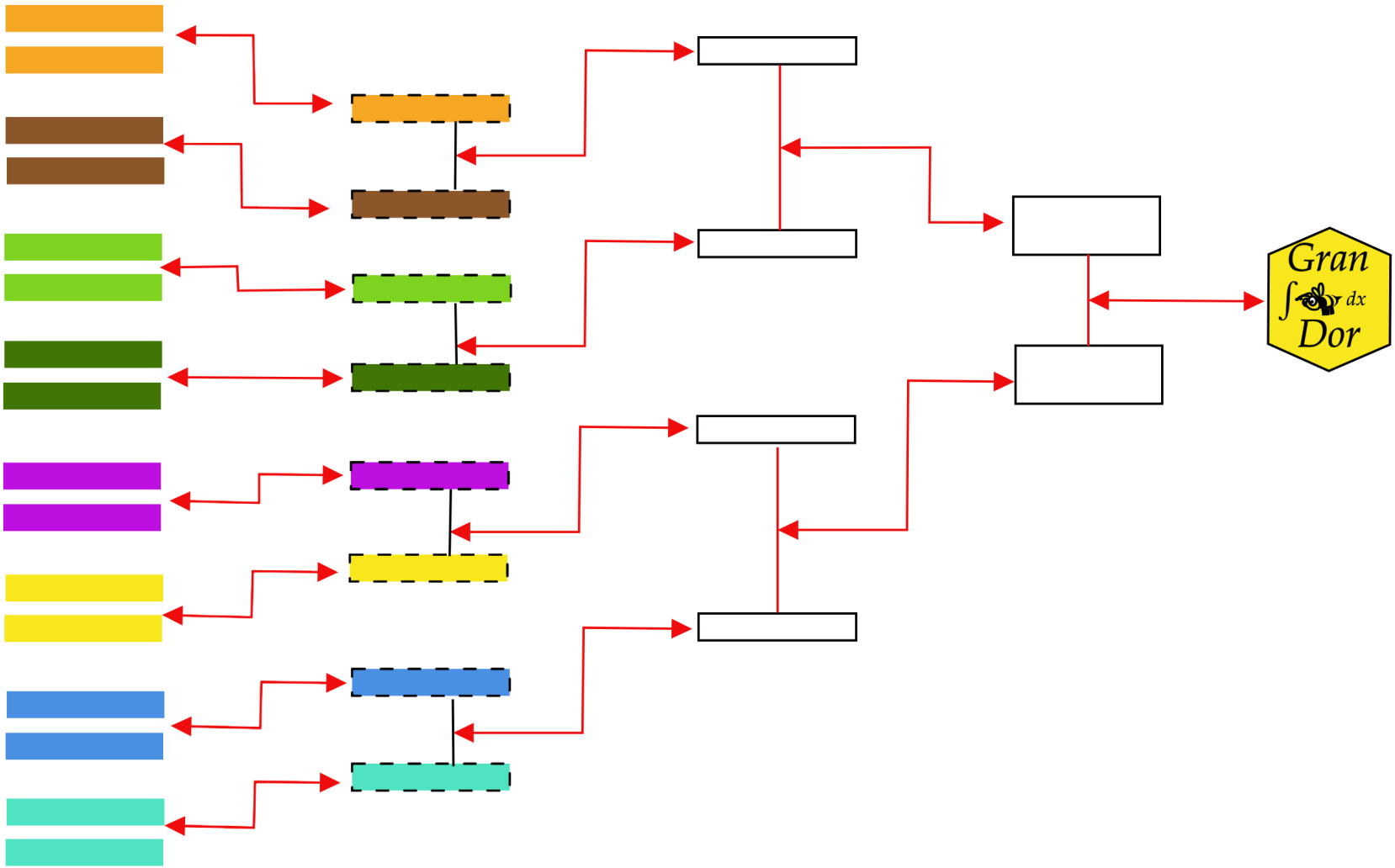
Integratión Bee Physics
Segunda etapa de clasificación



PRIMER TORNEO DE INTEGRALES
INTEGRATION BEE PHYSICS
CARRERA DE FISICA
UMSA

Reglas Integratión Bee Physic



Segunda Ronda

GRUPO 1

Integral #96 **Integratión Bee Physic**

$$\int \frac{1}{x^4\sqrt{1+x^2}}\,\mathrm{d}x$$

Respuesta Integral #96 | Integratión Bee Physic

$$\int \frac{1}{x^4\sqrt{1+x^2}}\,\mathrm{d}x =$$

$$-2(1+x^{-3/4})^{2/3} + C$$

Integral #2 Integratión Bee Physic

$$\int \frac{e^x - e^{2x}}{e^{3x} - 2e^{2x} + 2e^x - 1} \, \mathrm{d}x$$

Respuesta Integral #2 **Integratión Bee Physic**

$$\int \frac{e^x - e^{2x}}{e^{3x} - 2e^{2x} + 2e^x - 1} \mathrm{d}x =$$

$$\frac{2}{\sqrt{3}} \tan^{-1}\left[\frac{1 - 2e^x}{\sqrt{3}}\right] + C$$

Integral #86 **Integratión Bee Physic**

$$\int \frac{1}{x+\sqrt[3]{x}}\mathrm{d}x$$

$$\int \frac{1}{x+\sqrt[3]{x}}\,\mathrm{d}x =$$

$$\frac{3}{2}\ln(\sqrt[3]{x^2}+1)+C$$

GRUPO 2

Integral #7 Integratión Bee Physic

$$\int x \ln \sqrt{1+x^2}\, \mathrm{d}x$$

Respuesta Integral #7 **Integratión Bee Physic**

$$\int x \ln \sqrt{1+x^2}\, \mathrm{d}x =$$

$$\frac{1}{4}(1+x^2) - \frac{1}{2}(1+x^2)\ln\sqrt{1+x^2} + C$$

Integral #90 **Integratión Bee Physic**

$$\int_0^{2\pi} \tan(\cos x)\ \mathrm{d}x$$

$$\int_0^{2\pi} \tan(\cos x)\, \mathrm{d}x = 0$$

Integral #16 Integratión Bee Physic

$$\int [\ln x^x]^2 \, \mathrm{d}x$$

$$\int (\ln x^x)^2 \, \mathrm{d}x =$$

$$\frac{x^3}{3}\left[\ln^2 x - \frac{2\ln x}{3} - \frac{2}{9}\right] + C$$

GRUPO 3

Integral #41 **Integratión Bee Physic**

$$\int \frac{x^2+1}{x^4-x^2+1}\,\mathrm{d}x$$

$$\int \frac{x^2+1}{x^4-x^2+1}\,\mathrm{d}x =$$

$$\arctan\left(\frac{x^2-1}{x}\right)+C$$

Integral #47 **Integratión Bee Physic**

$$\int \frac{1}{x\sqrt{9x^2-1}}\,\mathrm{d}x$$

Respuesta Integral #47

Integratión Bee Physic

$$\int \frac{1}{x\sqrt{9x^2-1}}\,\mathrm{d}x =$$

$$\sec^{-1}(3x) + C$$

Integral #62 **Integratión Bee Physic**

$$\int \frac{x + \sin x - \cos x - 1}{x + e^x + \sin x} \mathrm{d}x$$

$$\int \frac{x + \sin x - \cos x - 1}{x + e^x + \sin x} \mathrm{d}x =$$

$$x - \ln|x + e^x + \sin x| + C$$

GRUPO 4

Integral #85 **Integratión Bee Physic**

$$\int_0^1 \sqrt{1-\sqrt{x}}\,\mathrm{d}x$$

Respuesta Integral #85

Integratión Bee Physic

$$\int_0^1 \sqrt{1-\sqrt{x}}\,\mathrm{d}x = \frac{8}{15}$$

Integral #1 Integratión Bee Physic

$$\int \frac{1}{\sqrt{1+e^x}}\,\mathrm{d}x$$

Respuesta Integral #1 **Integratión Bee Physic**

$$\int \frac{1}{\sqrt{1+e^x}}\,\mathrm{d}x =$$

$$\ln\left|\frac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1}\right| + C$$

Integral #89 Integratión Bee Physic

$$\int_0^e x^{\frac{1}{\ln x}} \, \mathrm{d}x$$

Respuesta Integral #89

Integratión Bee Physic

$$\int_0^e x^{\frac{1}{\ln x}}\,\mathrm{d}x = e^2$$

GRUPO 5

Integral #26 **Integratión Bee Physic**

$$\int \cos(4x)\cos(3x)\,\mathrm{d}x$$

$$\int \cos(4x)\cos(3x)\,\mathrm{d}x =$$

$$\frac{\sin x}{2} + \frac{\sin(7x)}{14} + C$$

Integral #45 **Integratión Bee Physic**

$$\int \frac{1}{x^2\sqrt{x^2+1}}\,\mathrm{d}x$$

$$\int \frac{1}{x^2\sqrt{x^2+1}}\,\mathrm{d}x =$$

$$-\sqrt{\frac{x^2+1}{x^2}}+C$$

Integral #30 **Integratión Bee Physic**

$$\int x \ln\left[1+\frac{1}{x}\right] \mathrm{d}x$$

$$\int x \ln\left[1+\frac{1}{x}\right] \mathrm{d}x =$$

$$\frac{x^2}{2}\ln\left(1+\frac{1}{x}\right)+\frac{1}{2}x-\frac{1}{2}\ln(1+x)+C$$

GRUPO 6

Integral #77 **Integratión Bee Physic**

$$\int \frac{1}{a^2 \sin^2 x + b^2 \cos^2 x} \mathrm{d}x$$

$$\int \frac{1}{a^2 \sin^2 x + b^2 \cos^2 x} \, \mathrm{d}x =$$

$$\frac{1}{b \cdot a} \arctan\left(\frac{a}{b} \tan x\right) + C$$

Integral #11 **Integratión Bee Physic**

$$\int \cos\sqrt{2x-1}\,\mathrm{d}x$$

$$\int \cos\sqrt{2x-1}\,\mathrm{d}x =$$

$$-\sqrt{2x-1}\sin\sqrt{2x-1}+\cos\sqrt{2x-1}+C$$

Integral #29 **Integratión Bee Physic**

$$\int_0^{\pi/2} \sin(x)\sin(2x)\sin(3x)\,\mathrm{d}x$$

$$\int_0^{\pi/2} \sin(x)\sin(2x)\sin(3x)\,\mathrm{d}x =$$

$$\frac{1}{6}$$

GRUPO 7

Integral #9 **Integratión Bee Physic**

$$\int \frac{\sin x}{e^x + \cos x + \sin x} \mathrm{d}x$$

Respuesta Integral #9 | Integratión Bee Physic

$$\int \frac{\sin x}{e^x + \cos x + \sin x}\,\mathrm{d}x =$$

$$\frac{x}{2} - \frac{1}{2}\ln\left|e^x + \cos x + \sin x\right| + C$$

Integral #79 Integratión Bee Physic

$$\int_0^{2\pi} \sin x \cdot \cos x \cdot \tan x \cdot \cot x \cdot \sec x \cdot \csc x \; \mathrm{d}x$$

Respuesta Integral #79

Integratión Bee Physic

$$\int_0^{2\pi} \sin x \cdot \cos x \cdot \tan x \cdot \cot x \cdot \sec x \cdot \csc x \, \mathrm{d}x =$$

$$2\pi$$

Integral #35 **Integratión Bee Physic**

$$\int \frac{1}{\sqrt{2}+\sqrt{1-x}+\sqrt{1+x}}\,\mathrm{d}x$$

Respuesta Integral #35

Integratión Bee Physic

$$\int \frac{1}{\sqrt{2}+\sqrt{1-x}+\sqrt{1+x}}\,\mathrm{d}x =$$

$$-\frac{\sqrt{2}}{2}\arcsin(x)+(1+x)^{1/2}+(1-x)^{1/2}+C$$